Reg. No. | | | | | | | | | | | [24MTEEM102]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY 2025

ENERGY CONVERSION ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Discuss the principles of overfeed and underfeed stoker coal firing system.	**1**	**2**	**10**
	b.	What is the difference between a crusher and a pulverizer? Explain any one type of pulverizer with a neat sketch.	**1**	**2**	**10**
2.	**a.**	Explain the working principle of diesel engine power plant with neat diagram.	**2**	**2**	**10**
	b.	Describe the working principles of open and closed gas turbine power plants. Include appropriate diagrams to illustrate each type.	**2**	**2**	**10**
3.	**a.**	How nuclear energy is used to generate electricity in a nuclear power plant? Explain with neat layout.	**3**	**2**	**10**
	b.	Discuss about current nuclear energy production scenario in India and in Global.	**3**	**3**	**10**
		OR			
4.	**a.**	With neat layout, discuss the concept of a breeder reactor and how it differs from traditional reactors in terms of fuel utilization.	**3**	**3**	**10**
	b.	Develop a strategy for the long-term management of nuclear waste.	**3**	**4**	**10**
5.	**a.**	Explain how tidal energy is harnessed using a tidal barrage and its advantages over conventional power generation.	**4**	**2**	**10**
	b.	Describe the working principle of a solar photovoltaic cell and its main construction components.	**4**	**2**	**10**
		OR			
6.	**a.**	Describe the typical layout of a hydroelectric power plant and identify the purpose of each major component.	**4**	**2**	**10**
	b.	Draw a line diagram of horizontal-axis wind turbine, label its components, and explain how energy is generated.	**4**	**3**	**10**

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7.	**a.**	Discuss the factors influencing site selection for hydro and thermal power station.	**5**	**2**	**10**
	b.	Describe the concept of a load curve and load duration curve. Explain their significance in power system planning.	**5**	**2**	**10**
		OR			
8.	**a.**	Explain the pollution control methods in thermal and nuclear power plants. Compare their advantages and limitations with respect to environmental safety.	**5**	**2**	**10**
	b.	A power station has a peak load of 30 MW. The maximum demands of individual loads are 15 MW, 10 MW, 5 MW, and 7 MW, respectively. If the station capacity is 40 MW and the annual load factor is 50%, calculate: i. The average load on the power station ii. The energy supplied per year iii. The demand factor iv. The diversity factor	**5**	**3**	**10**